

Analytical Solutions

So we have $-\phi'' + 4\phi = f$ on $0 \leq x \leq 1$ with PBC. Lets consider two cases:

Case 1: $f_1 = x$:

So first solve $-\phi'' + 4\phi = 0$. Notice that if we

let $\phi(r) = e^{rx}$, then, since $e^{rx''} = r^2 e^{rx}$, we have $r^2 e^{rx} - 4e^{rx} = 0 \Rightarrow r^2 - 4 = 0 \Rightarrow r = \pm 2$

Thus the homogeneous solution is

$$\phi_h(x) = C_1 e^{2x} + C_2 e^{-2x}$$

Now find the particular solution, try $ax + b$. Then, $\phi'' = 0$. Thus, $-0 + 4(ax + b) = x$. Eyeball it, then $4a = 1, b = 0 \Rightarrow a = \frac{1}{4}$. Thus:

$$\phi_r(x) = \frac{x}{4} \Rightarrow \phi(x) = C_1 e^{2x} + C_2 e^{-2x} + \frac{x}{4}$$

Apply PBC:

$$C_1 + C_2 = C_1 e^2 + C_2 e^{-2} + \frac{1}{4} \text{ and } 2C_1 - 2C_2 = 2C_1 e^2 - 2C_2 e^{-2} \text{ or } C_1(1 - e^2) = C_2(1 - e^{-2})$$

So we have

$$C_1(1 - e^2) + C_2(1 - e^{-2}) = \frac{1}{4} \text{ and } C_1(1 - e^2) - C_2(1 - e^{-2}) = 0$$

Let $a = C_1(1 - e^2)$ and $b = C_2(1 - e^{-2})$. Then,

$$a + b = \frac{1}{4} \text{ and } a - b = 0 \Rightarrow a = b \Rightarrow 2a = \frac{1}{4} \Rightarrow 2C_1(1 - e^2) = \frac{1}{4} \Rightarrow C_1 = \frac{1}{8(1 - e^2)}$$

and similarly,

$$b = \frac{1}{8} \Rightarrow C_2(1 - e^{-2}) = \frac{1}{8} \Rightarrow C_2 = \frac{1}{8(1 - e^{-2})}$$

Case 2: $f_2 = \sin(\pi x)$:

Eyeball it, consider $\phi_p = A \sin(\pi x) + B \cos(\pi x) \Rightarrow \phi_p'' = -A\pi^2 \sin(\pi x) - B\pi^2 \cos(\pi x) \Rightarrow$

$$\phi_p'' - 4\phi_p = (\pi^2 + 4)(A \sin(\pi x) + B \cos(\pi x)) \Rightarrow A = \frac{1}{\pi^2 + 4}, B = 0$$

Thus

$$\phi(x) = C_1 e^{2x} + C_2 e^{-2x} + \frac{\sin(\pi x)}{\pi^2 + 4}$$

Apply PBC:

$$C_1 + C_2 = C_1 e^2 + C_2 e^{-2} \Rightarrow (1 - e^2)C_1 + (1 - e^{-2})C_2 = 0$$

and

$$\begin{aligned} 2C_1 - 2C_2 + \frac{\pi}{\pi^2 + 4} &= 2C_1 e^2 - 2C_2 e^{-2} - \frac{\pi}{\pi^2 + 4} \Rightarrow 2C_1(1 - e^2) + 2C_2(e^{-2} - 1) = -\frac{2\pi}{\pi^2 + 4} \\ \Rightarrow C_1(1 - e^2) + C_2(e^{-2} - 1) &= -\frac{\pi}{\pi^2 + 4} \Rightarrow C_1(1 - e^2) - C_2(1 - e^{-2}) = -\frac{\pi}{\pi^2 + 4} \end{aligned}$$

Ok now solve for C_1, C_2 :

Let $a = C_1(1 - e^2), b = C_2(1 - e^{-2})$. So we have:

$$a + b = 0 \text{ and } a - b = -\frac{\pi}{\pi^2 + 4} \Rightarrow -b = a \Rightarrow 2a = \frac{-\pi}{\pi^2 + 4} \Rightarrow a = \frac{-\pi}{2(\pi^2 + 4)}$$

So,

$$C_1(1 - e^2) = \frac{-\pi}{2(\pi^2 + 4)} \Rightarrow C_1 = \frac{-\pi}{2(\pi^2 + 4)(1 - e^2)} = \frac{\pi}{2(\pi^2 + 4)(e^2 - 1)}$$

then,

$$a + b = 0 \Rightarrow b = \frac{\pi}{2(\pi^2 + 4)} \Rightarrow C_2(1 - e^{-2}) = \frac{\pi}{2(\pi^2 + 4)} \Rightarrow C_2 = \frac{\pi}{2(\pi^2 + 4)(1 - e^{-2})}$$

Numerical Solutions:

We now solve numerically using a finite difference scheme:

Let $N = 2^q$. Then, we have $h = \frac{1}{N}$, $x_j = jh$, $0 \leq j \leq N - 1$.

Then use what was described in notes:

$$\phi_j'' \approx \frac{\phi_{j+1} - 2\phi_j + \phi_{j-1}}{h^2} \Rightarrow -\frac{\phi_{j+1} - 2\phi_j + \phi_{j-1}}{h^2} + 4\phi_j \approx f_j$$

Notice

$$f_j \approx (4 + \frac{2}{h^2})\phi_j + -\frac{1}{h^2}(\phi_{j+1} + \phi_{j-1}) \Rightarrow h^2 f_j \approx (4h^2 + 2)\phi_j - \phi_{j-1} - \phi_{j+1}$$

Then we have the matrix:

$$A = \frac{1}{h^2} \begin{bmatrix} 2 + 4h^2 & -1 & \cdots & -1 \\ -1 & 2 + 4h^2 & & -1 \\ & & \ddots & \\ -1 & \cdots & & 2 + 4h^2 \end{bmatrix}$$

Then we have $Au = f$, solve for u . Observe we have that by the notes, $AF_N^* = F_N^*D$, $D = \text{diag}(\lambda_0, \dots, \lambda_{N-1})$, $\lambda_n = 4 + \frac{4\sin^2(\pi nh)}{h^2}$. Thus we have $A = F_N^*DF_N$.

So we have $u = A^{-1}f = F_N^*D^{-1}F_N f$.

Note: I omitted the numerical solutions for the other two for brevity. They follow roughly the same idea; the point is that you end up with eigenvalues that you can then use in Python/MATLAB as part of the $F_N^*D^{-1}F_N$ form to get u .

For more precise explanations and/or notes on these numerical methods, I'd recommend checking out Robert Krasny's page and looking at his lecture notes under Math 671b.

Plotting errors and rates of convergence:

For brevity, we'll only include finite difference and Green's function. Last I checked, I got the following numbers:

$f(x) = x$, method: finite difference

q	N	h	error	p
2	4	0.25	≈ 0.04003	—
3	8	0.125	≈ 0.02038	0.974
4	16	0.0625	≈ 0.01024	0.993
5	32	0.03125	≈ 0.00512	0.998
6	64	0.015625	≈ 0.00256	1.000
7	128	0.0078125	≈ 0.00128	1.000

$f(x) = \sin(\pi x)$, method: finite difference

q	N	h	error	p
2	4	0.25	≈ 0.01323	—
3	8	0.125	≈ 0.00338	1.967
4	16	0.0625	≈ 0.00085	1.992
5	32	0.03125	≈ 0.00021	1.998
6	64	0.015625	$\approx 5.3281 \times 10^{-5}$	1.999
7	128	0.0078125	$\approx 1.3321 \times 10^{-5}$	2.000

$f(x) = x$, method: Green's function

q	N	h	error	p
2	4	0.25	≈ 0.03843	—
3	8	0.125	≈ 0.01986	0.952
4	16	0.0625	≈ 0.01009	0.977
5	32	0.03125	≈ 0.00508	0.988
6	64	0.015625	≈ 0.00255	0.994
7	128	0.0078125	≈ 0.00127	0.997

$f(x) = \sin(\pi x)$, method: Green's function

q	N	h	error	p
2	4	0.25	≈ 0.01071	—
3	8	0.125	≈ 0.00268	1.997
4	16	0.0625	≈ 0.00067	1.999
5	32	0.03125	≈ 0.00016	2.000
6	64	0.015625	$\approx 4.1961 \times 10^{-5}$	2.000
7	128	0.0078125	$\approx 1.0490 \times 10^{-5}$	2.000

The main ordeal about this is that we see that across all methods, the identity function has a convergence rate of $O(h)$, not $O(h^2)$, though this is just because the identity function is not periodic, causing it to have a discontinuity when extended periodically. Also, the results are not so similar for the pseudospectral method, since it exploits smoothness that neither function has to a sufficient degree.